

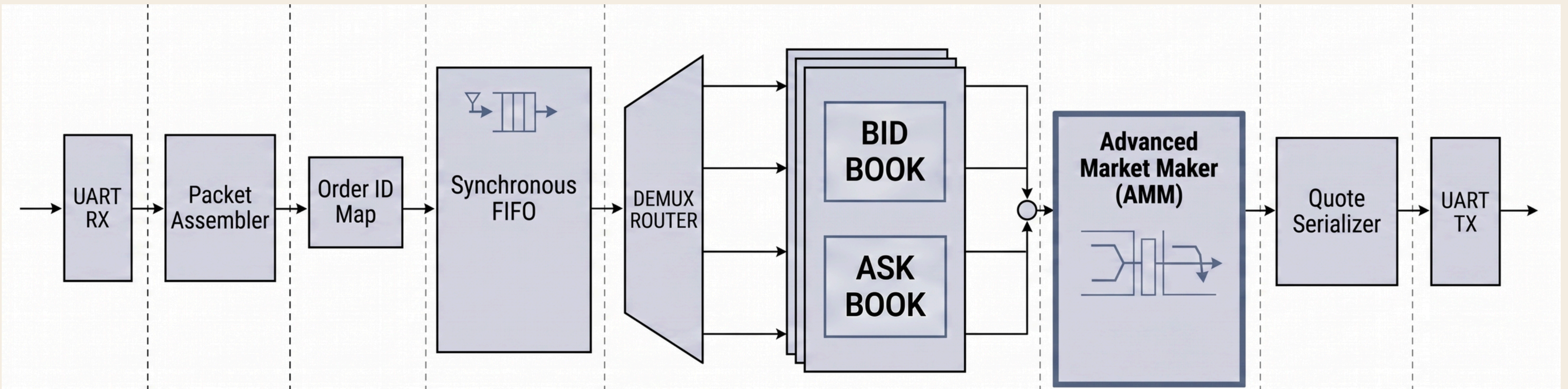
LATENCY-DETERMINISTIC FPGA BASED HIGH-FREQUENCY TRADING (HFT) SYSTEM

MiniProject in DSD

Sachin V Nagaraddi (241EC246)

Vamshikrishna V Bidari (241EC258)

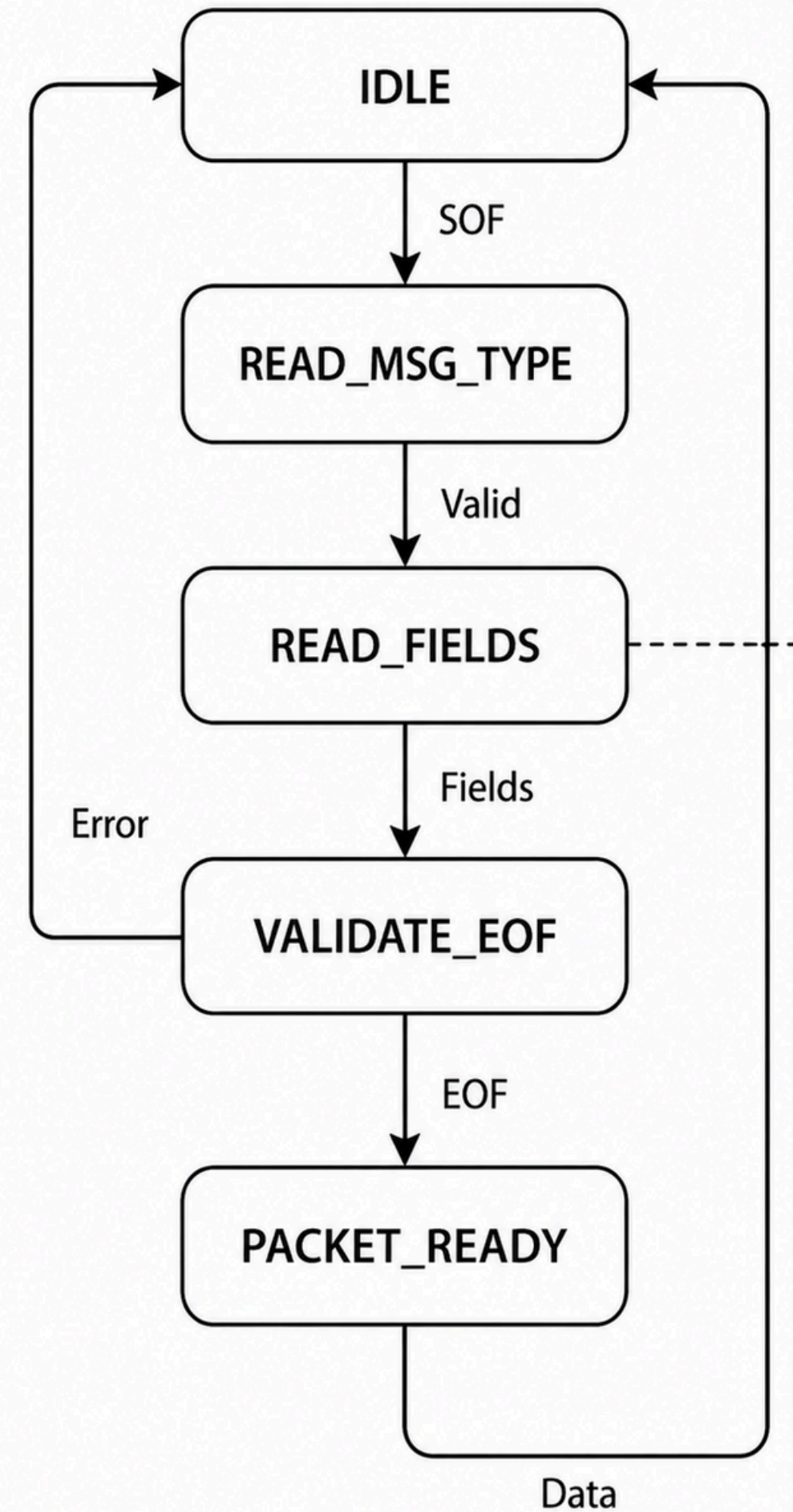
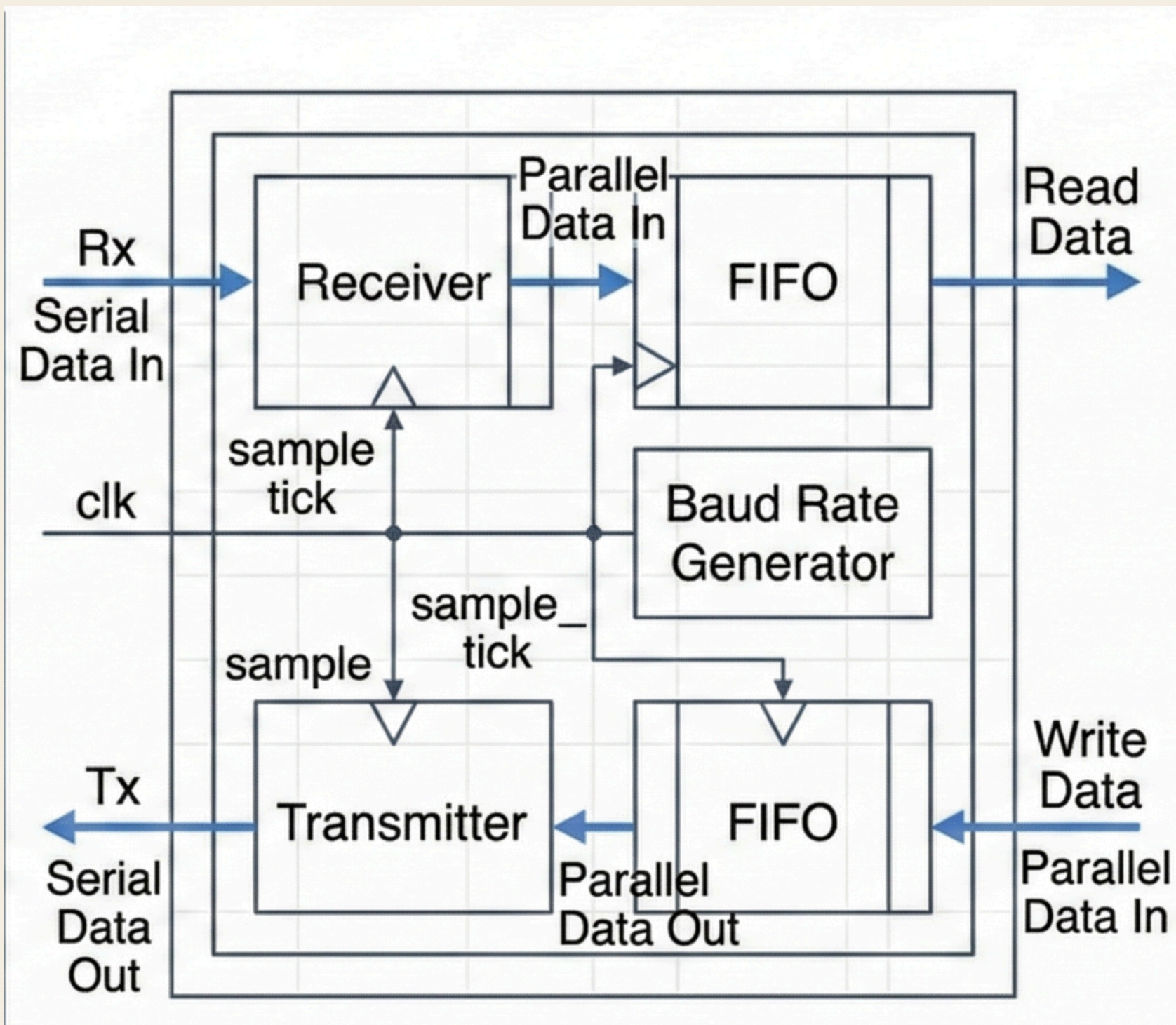
SYSTEM ARCHITECTURE



HOST PC → UART RX → **FPGA CORE** → UART TX → HOST PC

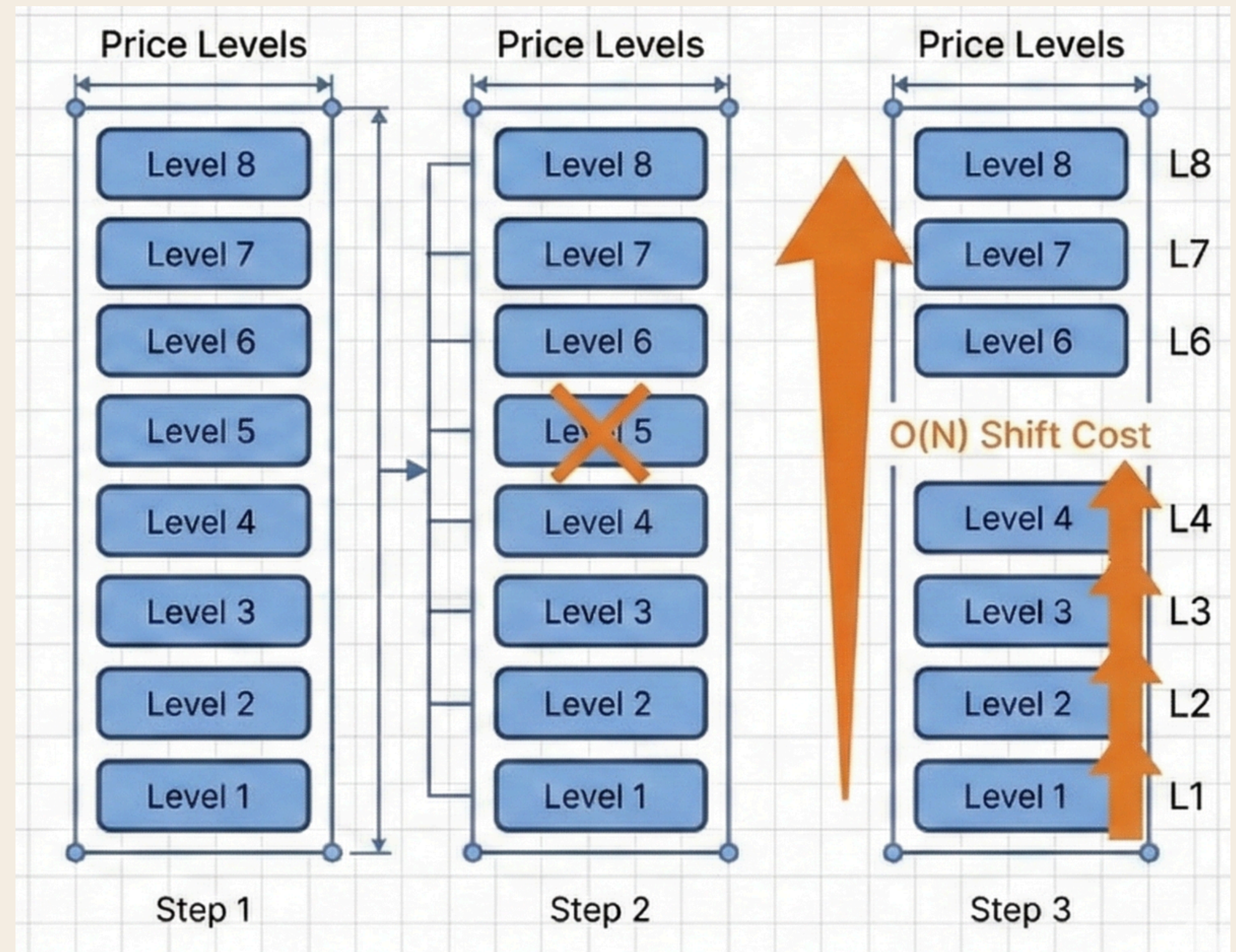
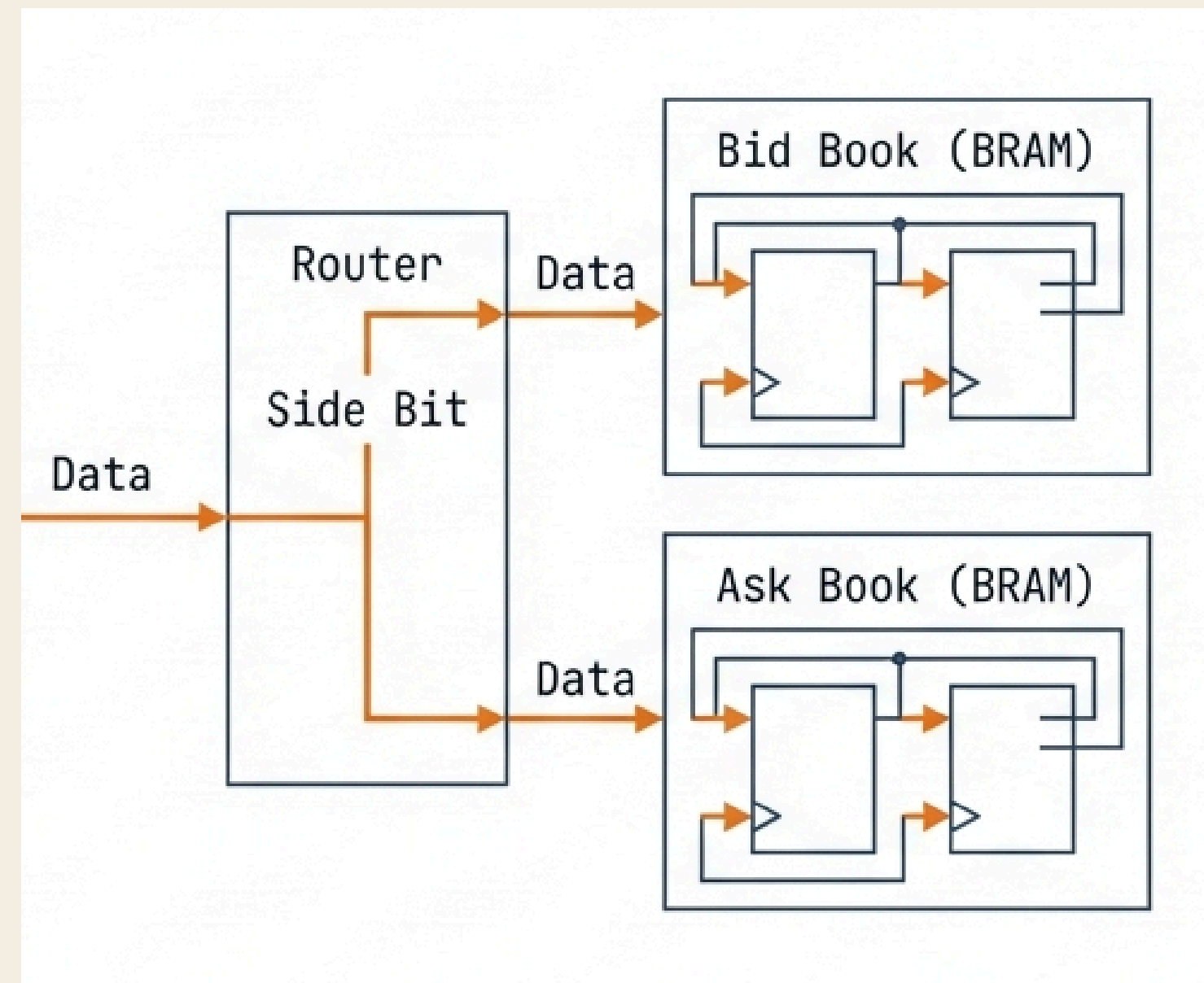
RECAP

UART CORE & PARSER FSM



RECAP

ROUTING, ORDERBOOK & BRAM SHIFT COST

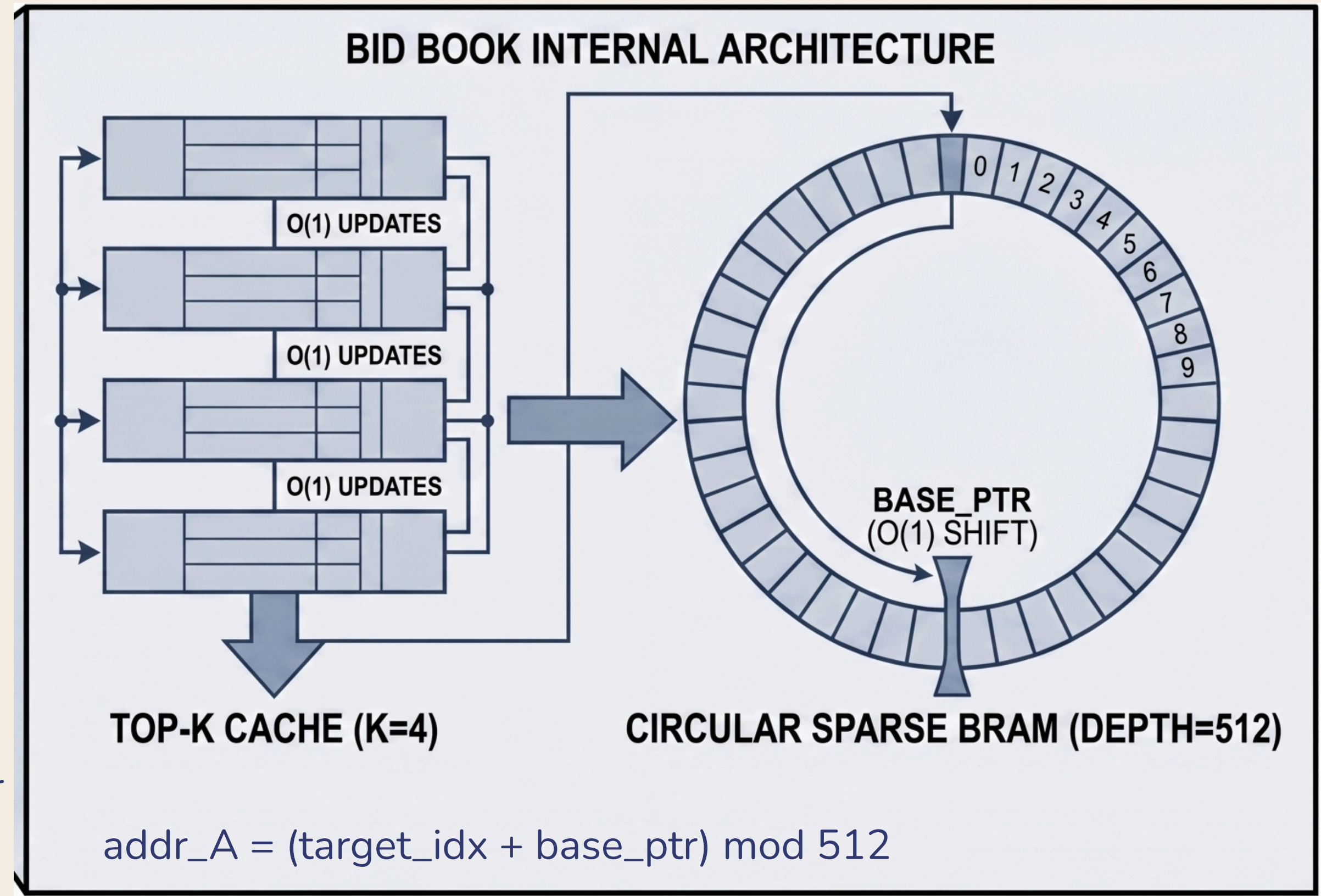


HYBRID ORDER BOOK ENGINE

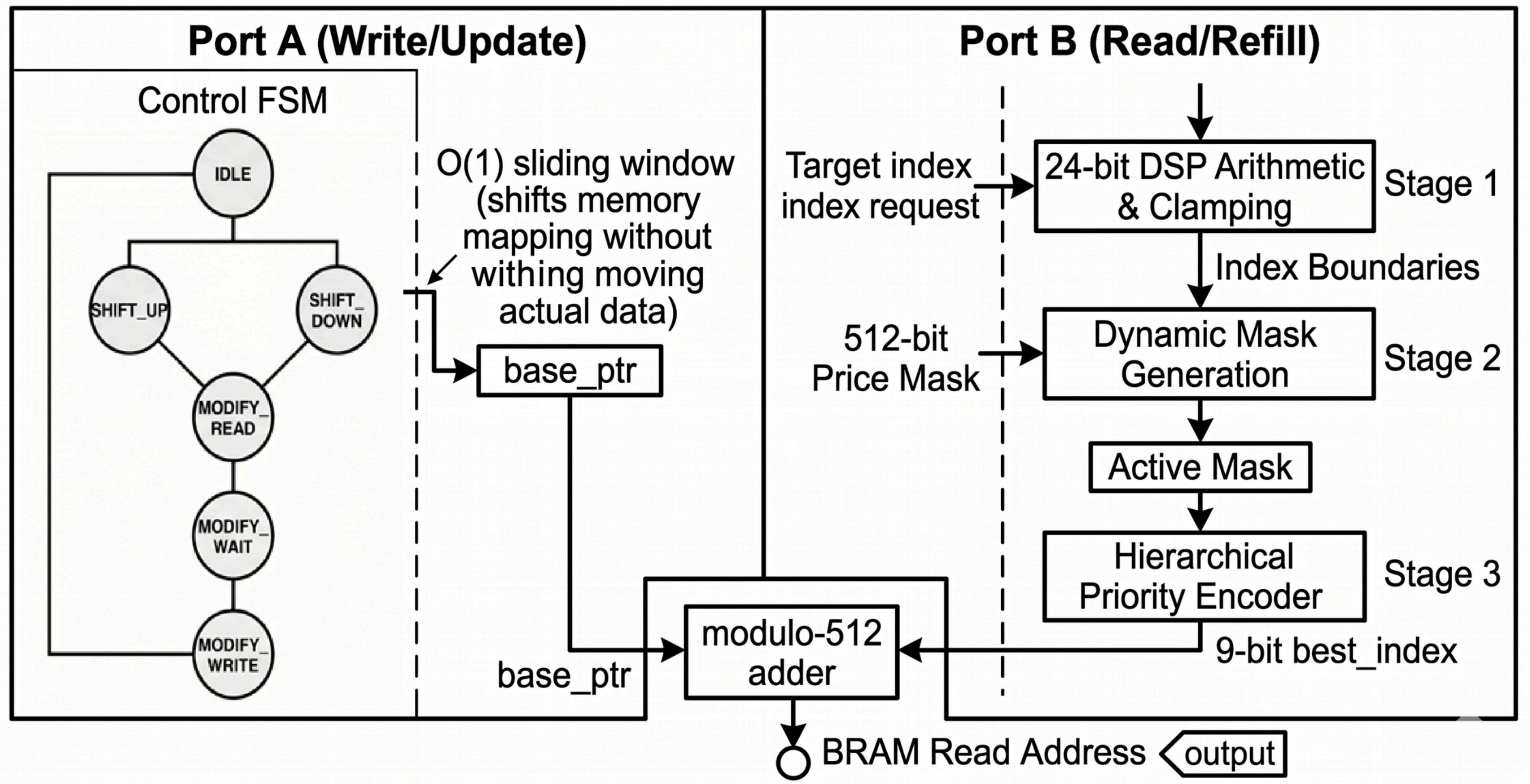
PRICE REPRESENTATION

- Scaling: $\times 100$ Integer Conversion
- Tick Size: 1 Tick = ₹0.01
- Mapping Example: ₹100.00 $\rightarrow$ 10000
- (Stored as 24-bit Integer)
- Eliminates floating-point latency
- Coverage: Price-center $\pm$ ₹2.56

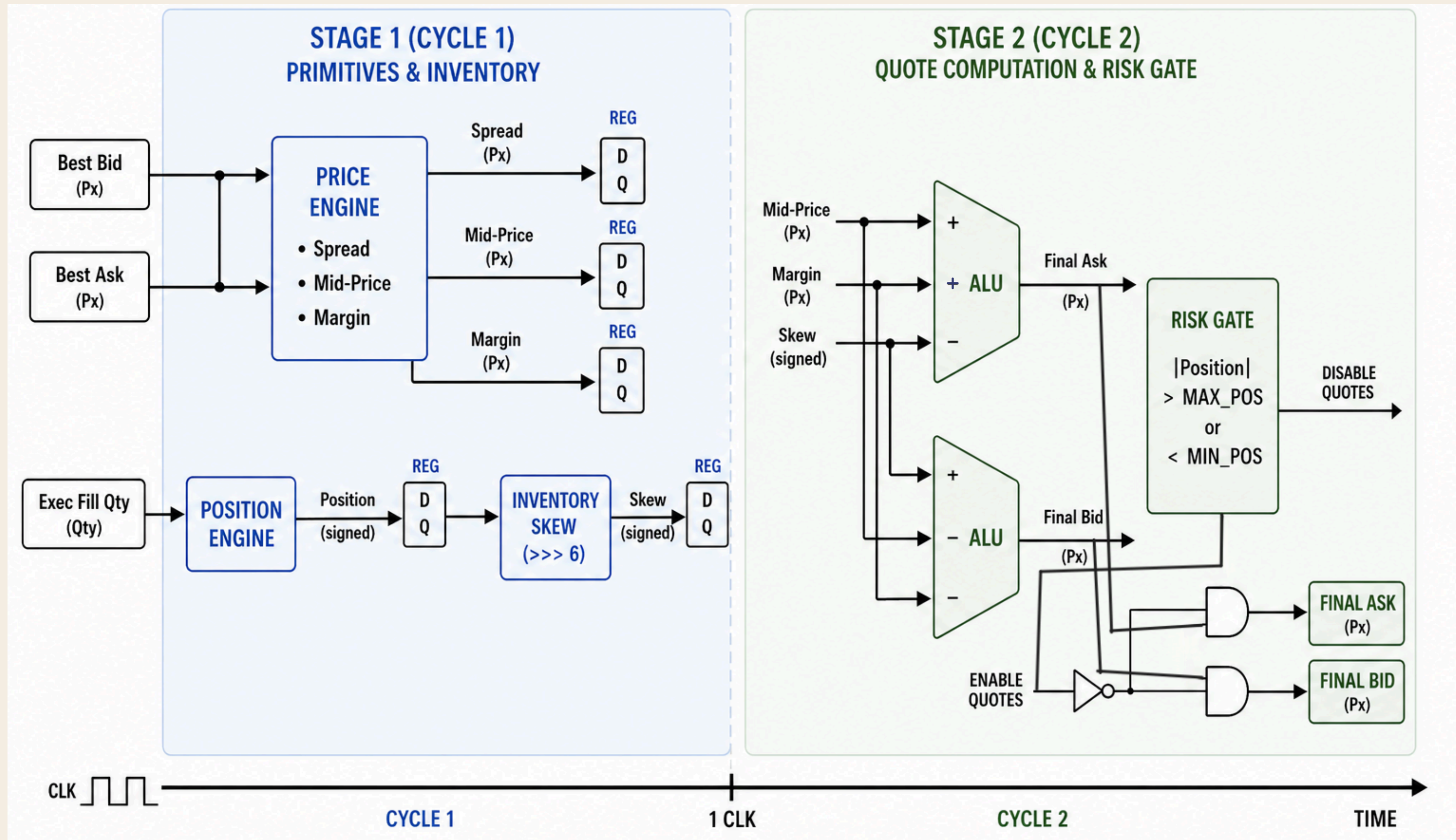
$\text{price_diff} = \text{price_in} - \text{price_center}$
 $\text{target_idx} = \text{price_diff} + 256$



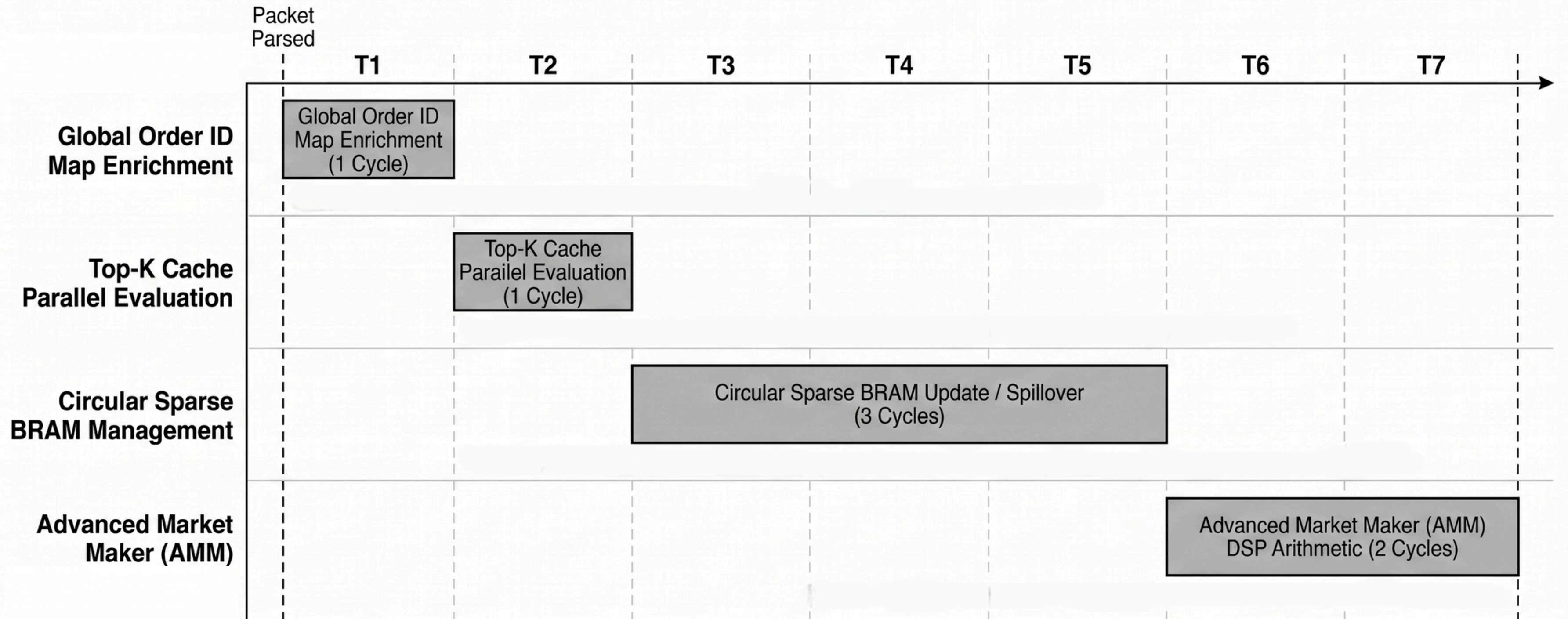
CIRCULAR SPARSE BRAM



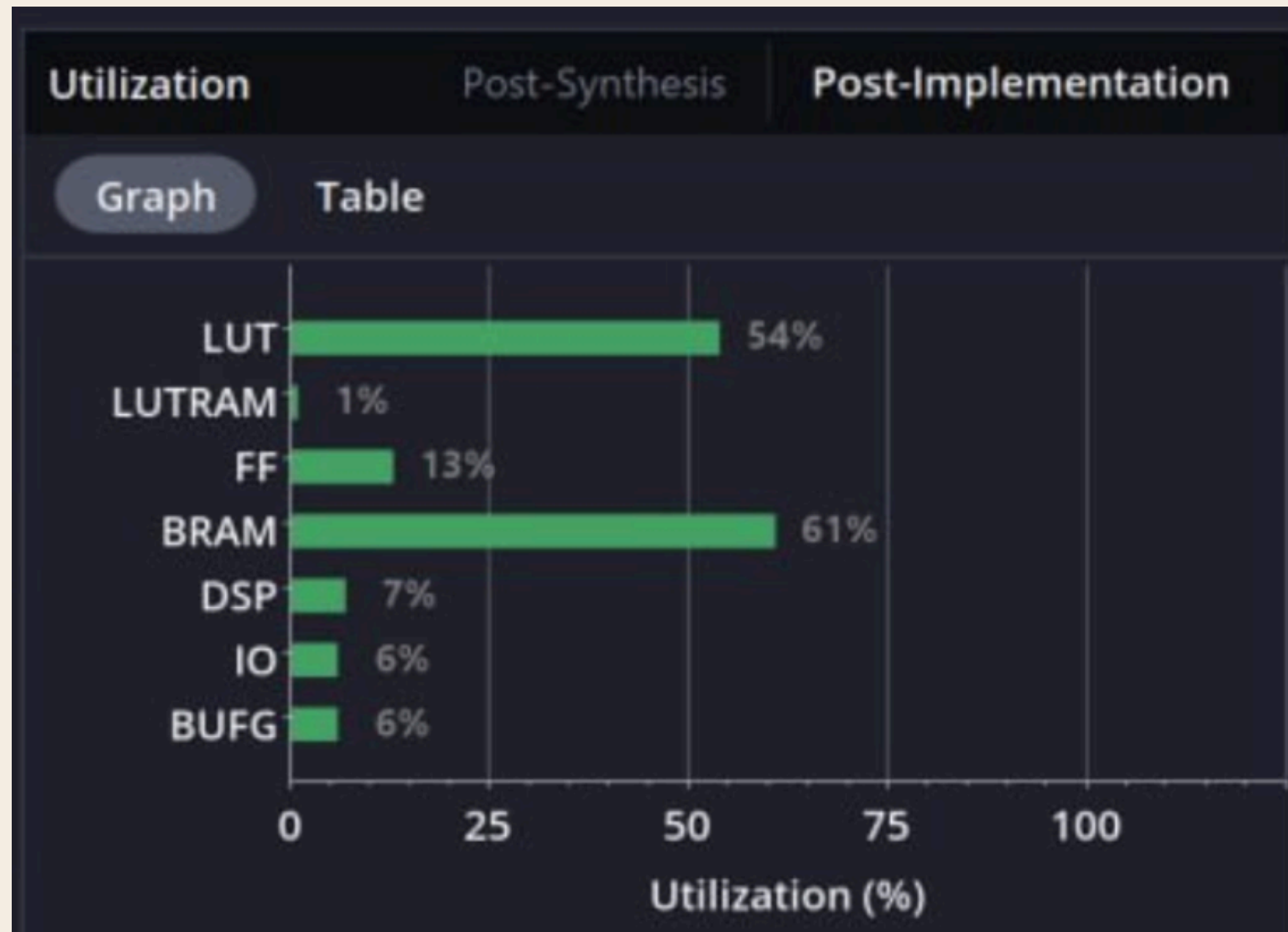
ADVANCED MARKET MAKER



EXECUTION TIMELINE



RESOURCE USAGE & TIMING CLOSURE



4-stock OrderBook System

Worst Negative Slack (WNS)	0.038 ns
Total Negative Slack (TNS)	0 ns
Number of Failing Endpoints	0

Single-stock OrderBook System:

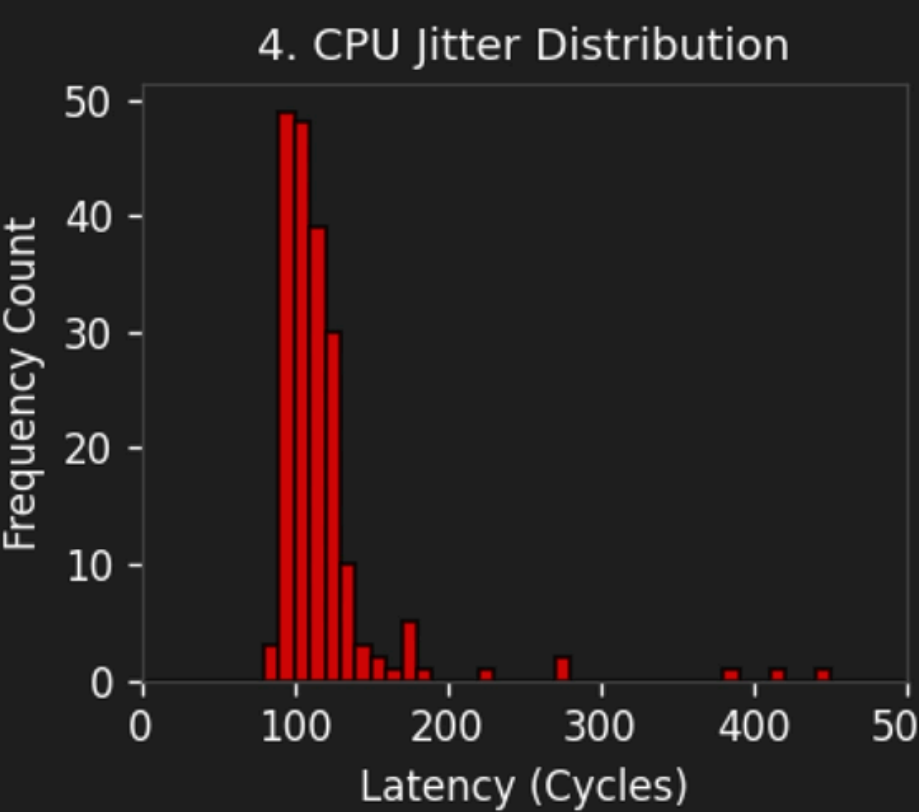
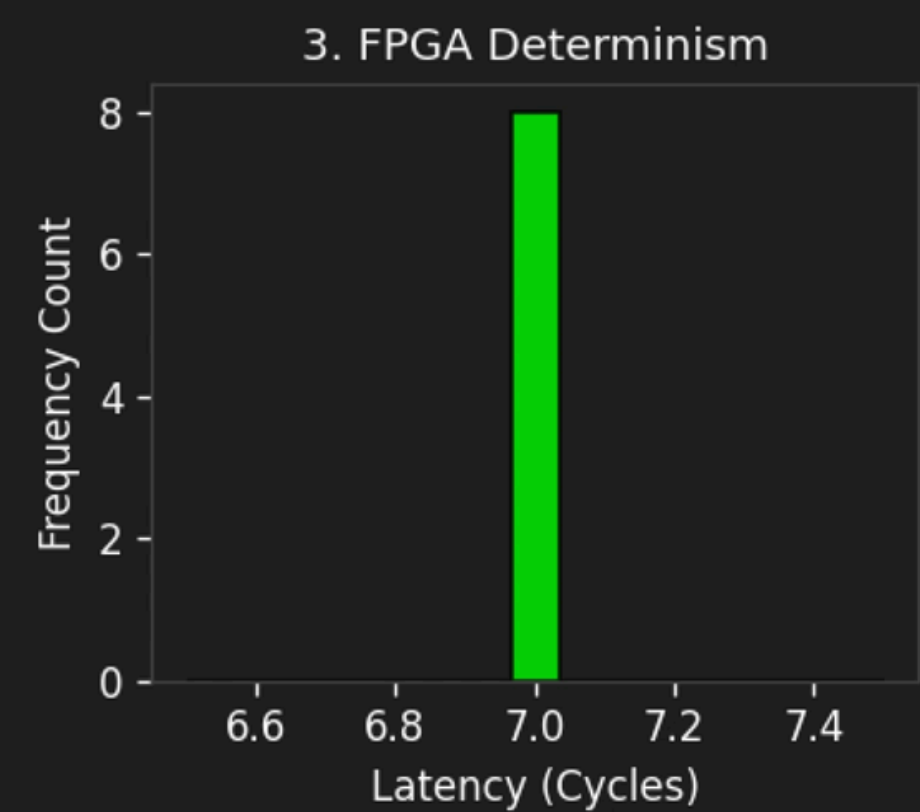
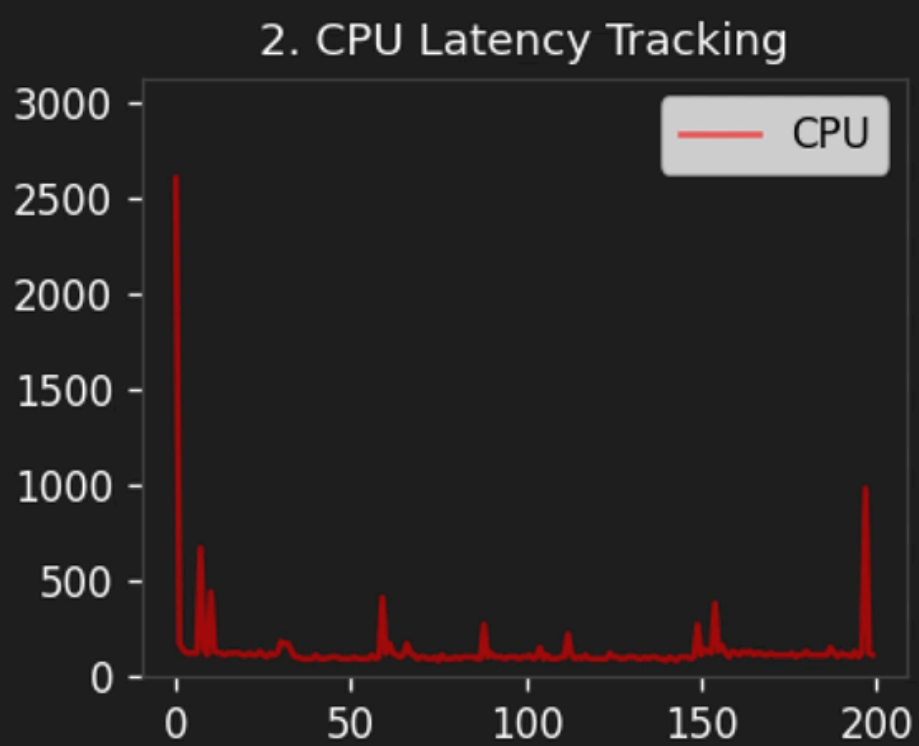
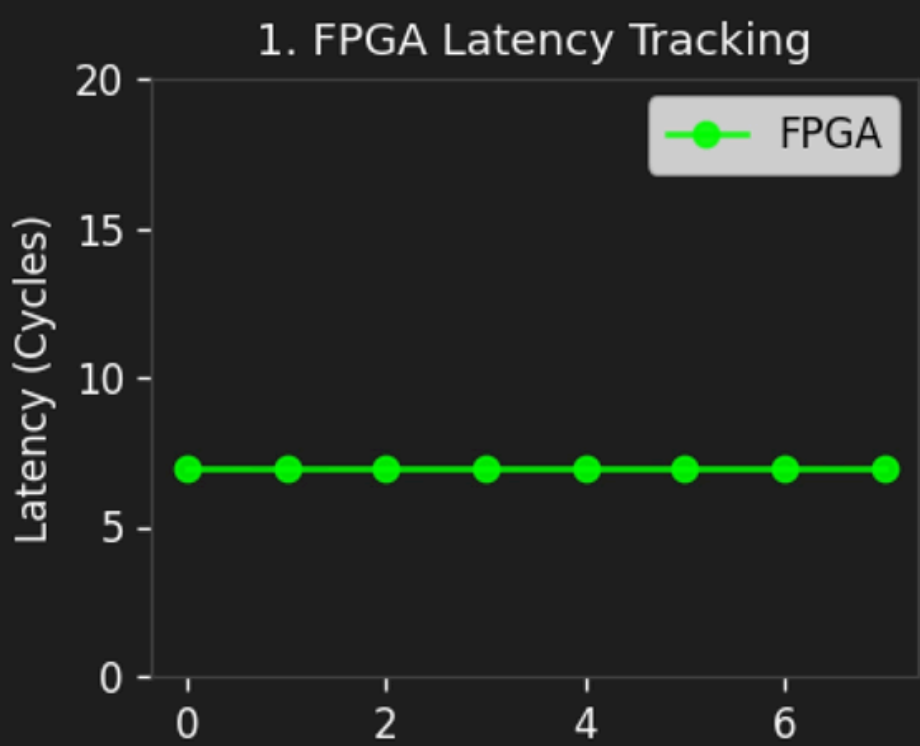
- LUTs = 16.11%
- FFs = 5.32 %
- BRAM = 59.64 %
- DSP = 1.82 %

HFT ARCHITECTURE BENCHMARK

Test Scenario: Data Pattern 5

▶ START TEST

Running...



LIVE METRICS

--- DASHBOARD METRICS ---

Orders Processed:
Total sent : 200 (CPU)
FPGA : 8

Average Latency:
CPU Model : 133.95 cycles
FPGA Model : 7.00 cycles

Max Jitter / Peak Latency:
CPU Peak : 2600 cycles
FPGA Peak : 7 cycles

LIVE QUOTES FEED

TSLA	BUY	99.95	Q:100	7c
AAPL	BUY	100.08	Q:100	7c
TSLA	BUY	100.08	Q:100	7c
AAPL	BUY	99.94	Q:100	7c
GOOG	BUY	100.07	Q:100	7c
AAPL	SELL	100.08	Q:200	7c
MSFT	SELL	100.10	Q:200	7c
MSFT	SELL	100.08	Q:200	7c

CPU v/s FPGA

REFERENCES

- Endrias, Tony, and Natnael, “An HFT (High Frequency Trading) Accelerator,” Massachusetts Institute of Technology (MIT), 6.111 Final Project Report, Fall 2019.
- National Stock Exchange of India (NSE), “Real-time Capital Market / Currency Derivatives Tick-By-Tick (MTBT) Protocol,” Version 6.4
- D. J. Marion (FPGADude), “Digital-Design/FPGA Projects/UART,” GitHub.
- C. Leber, B. Geib, and H. Litz, “High frequency trading acceleration using FPGAs,” in 2011 21st International Conference on Field Programmable Logic and Applications, Chania, Greece, 2011, pp. 317-322.

THANK YOU!